1 Team Problems

Problem 1. Compute the number of non-empty subsets S of $\{1, 2, 3, \ldots, 20\}$ for which $|S| \cdot \max\{S\} = 18$. (Note: $|S|$ is the number of elements of the set S.)

Problem 2. A class of 218 students takes a test. Each student's score is an integer from 0 to 100, inclusive. Compute the greatest possible difference between the mean and the median scores.

Problem 3. Regular hexagon $RANGES$ has side length 6. Pentagon $RANGE$ is revolved 360° about the line containing $\overline{RE}$ to obtain a solid. The volume of the solid is $k \cdot \pi$. Compute k.

Problem 4. A fair 12-sided die has faces numbered 1 through 12. The die is rolled twice, and the results of the two rolls are x and y, respectively. Given that $\tan(2\theta) = \frac{x}{y}$ for some θ with $0 < \theta < \frac{\pi}{2}$, compute the probability that $\tan\theta$ is rational.

Problem 5. The absolute values of the five complex roots of $z^5 - 5z^2 + 53 = 0$ all lie between the positive integers a and b, where $a < b$ and $b - a$ is minimal. Compute the ordered pair (a, b).

Problem 6. Let S be the set of points (x, y) whose coordinates satisfy the system of equations:

$$\lfloor x \rfloor \cdot \lceil y \rceil = 20$$
$$\lceil x \rceil \cdot \lfloor y \rfloor = 18.$$

Compute the least upper bound of the set of distances between points in S.

Problem 7. Compute the least integer $d > 0$ for which there exist distinct lattice points A, B, and C in the coordinate plane, each exactly $\sqrt{d}$ units from the origin, satisfying $\csc(\angle ABC) > 2018$.

Problem 8. Compute the number of unordered collections of three integer-area rectangles such that the three rectangles can be assembled without overlap to form one 3×5 rectangle. (For example, one such collection contains one 3×3 and two 1×3 rectangles, and another such collection contains one 3×3 and two 2×1.5 rectangles. The latter collection is equivalent to the collection of two 1.5×2 rectangles and one 3×3 rectangle.)

Problem 9. Let Γ be a circle with diameter $\overline{XY}$ and center O, and let γ be a circle with diameter $\overline{OY}$. Circle ω_1 passes through Y and intersects Γ and γ again at A and B, respectively. Circle ω_2 also passes through Y and intersects Γ and γ again at D and C, respectively. Given that $AB = 1$, $BC = 4$, $CD = 2$, and $AD = 7$, compute the sum of the areas of ω_1 and ω_2.

Problem 10. In the number puzzle below, clues are given for the four rows, each of which contains a four-digit number. Cells inside a region bounded by bold lines must all contain the same digit, and each of the eight regions contains a different digit. The variables in the clues are all positive integers. Complete the number puzzle.

1: $4^a + 13^b + 14^c$

2: $5^p + 13^q + 17^r$

3: $4^x + 5^y + 31^z$

4: the average of the other 3 rows

